

The DevOps Troubleshooting Handbook



By DevOps Shack

DevOps Shack

The DevOps Troubleshooting Handbook

Introduction

In the fast-paced world of software development and IT operations, DevOps has emerged as a critical approach to bridging the gap between development and operations teams. By fostering collaboration, automation, and continuous delivery, DevOps empowers organizations to deliver high-quality software faster and more reliably. However, with great power comes great complexity. As the DevOps ecosystem grows to encompass a multitude of tools and technologies—such as Git, Jenkins, Docker, Kubernetes, Ansible, Terraform, and monitoring stacks—teams face an increasing number of challenges, errors, and roadblocks.

Whether you're a seasoned DevOps engineer or just starting your journey, troubleshooting is an inevitable and essential part of the role. From configuration issues and permission errors to deployment failures and system-level bugs, these problems can disrupt workflows, impact productivity, and even delay critical software releases. Understanding how to identify, analyze, and resolve these issues is not just a skill—it's a necessity.

This guide, "**DevOps Troubleshooting: A Comprehensive Guide**," is designed to be your go-to resource for addressing the most common and challenging errors across a wide array of DevOps tools. Structured tool by tool, the book explores the top 10 errors for each major technology, providing detailed explanations, root causes, and actionable solutions to help you overcome these challenges with confidence.

By learning from these errors, you'll not only become adept at troubleshooting but also gain insights into building more resilient, automated, and efficient systems. As you dive into this book, remember: every error is an opportunity to learn, innovate, and enhance your expertise in the ever-evolving world of DevOps.

Welcome to the journey of mastering DevOps troubleshooting—let's get started!

Table of Contents

1. Git Errors

- 1.1 "fatal: not a git repository (or any of the parent directories): .git"
 - 1.2 "error: failed to push some refs"
 - 1.3 "Permission denied (publickey)"
 - 1.4 "Merge conflict in [file]"
 - 1.5 "Detached HEAD state"
 - 1.6 "fatal: remote origin already exists"
 - 1.7 "Large file exceeds limit"
 - 1.8 "Submodule update failed"
 - 1.9 "fatal: cannot lock ref"
 - 1.10 "Untracked files prevent switching branches"
-

2. Jenkins Errors

- 2.1 "Jenkins service not starting"
 - 2.2 "Build stuck in the queue"
 - 2.3 "Plugins fail to load"
 - 2.4 "Pipeline script syntax error"
 - 2.5 "Build fails due to missing environment variables"
 - 2.6 "Unauthorized webhook trigger"
 - 2.7 "Job not found after restart"
 - 2.8 "Out of disk space"
 - 2.9 "Build artifacts not archiving"
 - 2.10 "Node disconnected unexpectedly"
-

3. Docker Errors

- 3.1 "Cannot connect to the Docker daemon"
- 3.2 "Port is already in use"
- 3.3 "No space left on device"
- 3.4 "ImagePullBackOff" in Kubernetes
- 3.5 "Permission denied on bind mount"
- 3.6 "Container restart loop"

-
- 3.7 "Failed to build Docker image"
 - 3.8 "Error response from daemon: conflict"
 - 3.9 "Docker network not reachable"
 - 3.10 "Volume mount not working"
-

4. Kubernetes Errors

- 4.1 "ImagePullBackOff"
 - 4.2 "CrashLoopBackOff"
 - 4.3 "Node Not Ready"
 - 4.4 "PersistentVolumeClaim Pending"
 - 4.5 "Pod is stuck in Pending state"
 - 4.6 "RBAC: Access Denied"
 - 4.7 Service Unreachable
 - 4.8 "Resource Quota Exceeded"
 - 4.9 "Evicted Pods"
 - 4.10 Deployment Rollout Fails
-

5. Ansible Errors

- 5.1 "Host unreachable"
 - 5.2 "Syntax error in playbook"
 - 5.3 "Undefined variable"
 - 5.4 "Command not found on target node"
 - 5.5 "Permission denied"
 - 5.6 Module not found
 - 5.7 "Failed to find group_vars"
 - 5.8 Playbook runs indefinitely
 - 5.9 "Handler not triggered"
 - 5.10 Dynamic inventory script failure
-

6. Terraform Errors

- 6.1 "Provider plugin not found"
- 6.2 "State file lock error"
- 6.3 "Error acquiring the state lock"
- 6.4 Resource already exists

-
- 6.5 "Plan does not match changes"
 - 6.6 "Invalid index" in Terraform output
 - 6.7 "Module not found"
 - 6.8 Authentication failure
 - 6.9 "Timeout waiting for resource"
 - 6.10 "Remote backend configuration error"
-

7. Prometheus Errors

- 7.1 "No targets found"
 - 7.2 "Prometheus not scraping data"
 - 7.3 "High cardinality in metrics"
 - 7.4 "Prometheus service not starting"
 - 7.5 "Query taking too long"
 - 7.6 "Prometheus out of storage space"
 - 7.7 "Prometheus alert not firing"
 - 7.8 "Prometheus crash due to OOM (Out of Memory)"
 - 7.9 "Scraped data is incomplete"
 - 7.10 "Failed to reload configuration"
-

8. ELK Stack Errors

- 8.1 Elasticsearch: "Cluster health is red"
 - 8.2 Elasticsearch: "Java heap space error"
 - 8.3 Logstash: "Pipeline aborted due to error"
 - 8.4 Logstash: "Connection refused to Elasticsearch"
 - 8.5 Kibana: "Kibana server is not ready yet"
 - 8.6 "Index pattern not found in Kibana"
 - 8.7 "Logstash not processing logs"
 - 8.8 Elasticsearch: "Index not found"
 - 8.9 Kibana: "Dashboard is empty"
 - 8.10 Logstash: "Filebeat logs not received"
-

9. AWS DevOps Tools Errors

- 9.1 EC2: "Instance not reachable"
- 9.2 S3: "Access Denied"

-
- 9.3 CodePipeline: "Failed to deploy"
 - 9.4 RDS: "Cannot connect to the database"
 - 9.5 CloudFormation: "Stack creation failed"
 - 9.6 IAM: "Policy not authorized"
 - 9.7 ECS: "Task failed to start"
 - 9.8 Lambda: "Execution failed"
 - 9.9 Route 53: "DNS record not resolving"
 - 9.10 CloudWatch: "Metrics not visible"
-

10. Azure DevOps Tools Errors

- 10.1 Pipelines: "Build agent unavailable"
 - 10.2 "Resource group deployment failed"
 - 10.3 "Release failed in Azure DevOps"
 - 10.4 Repos: "Merge conflict during pull request"
 - 10.5 "Pipeline YAML syntax error"
 - 10.6 "Artifacts not found"
 - 10.7 "Access denied to the Azure subscription"
 - 10.8 "Cannot connect to Azure Kubernetes Service (AKS)"
 - 10.9 Boards: "Work item cannot be updated"
 - 10.10 "Failed to publish test results"
-

11. CI/CD Pipeline Errors

- 11.1 "Pipeline stuck in pending state"
 - 11.2 "Build fails due to dependency issues"
 - 11.3 "Environment variable not found"
 - 11.4 "Permission denied during deployment"
 - 11.5 "Timeout in build or deployment stage"
 - 11.6 "Pipeline fails only on specific branches"
 - 11.7 "Docker image not found in the pipeline"
 - 11.8 "Failed to upload artifacts"
 - 11.9 "Parallel jobs failing intermittently"
 - 11.10 "Webhook not triggering pipeline"
-

12. Monitoring Tools Errors

- 12.1 "Metrics not visible in Grafana"
- 12.2 "Prometheus data missing from dashboards"
- 12.3 "Logs not visible in ELK Stack"
- 12.4 "High load on monitoring servers"
- 12.5 "Alerts not firing in Prometheus"
- 12.6 "Outdated Grafana dashboards"
- 12.7 "Log ingestion delay in ELK Stack"
- 12.8 "Scraped data is incomplete"
- 12.9 "Too many false-positive alerts"
- 12.10 "Dashboard panels showing no data"

1. Git Errors

1.1 "fatal: not a git repository (or any of the parent directories): .git"

- **Cause:** The command is executed outside a Git repository or the repository is corrupted.
- **Solution:**
 1. Verify you're in the correct directory: `pwd`.
 2. Navigate to the repository: `cd <repository-path>`.
 3. If missing, reinitialize the repository: `git init`.

1.2 "error: failed to push some refs"

- **Cause:** Local branch is out of sync with the remote branch.
- **Solution:**
 1. Pull latest changes with rebase: `git pull --rebase origin <branch>`.
 2. Resolve conflicts if prompted, then commit and push.

1.3 "Permission denied (publickey)"

- **Cause:** SSH key is missing or not recognized by the remote repository.
- **Solution:**
 1. Generate a key pair: `ssh-keygen -t rsa -b 4096`.
 2. Add the private key to the agent: `ssh-add ~/.ssh/id_rsa`.
 3. Add the public key to the repository settings.

1.4 "Merge conflict in [file]"

- **Cause:** Simultaneous changes to the same part of a file.
- **Solution:**
 1. Open the conflicting file to resolve changes.
 2. Use markers like `<<<<<<<` and `>>>>>>>` to identify conflicts.
 3. After resolution, commit: `git add <file> && git commit`.

1.5 "Detached HEAD state"

- **Cause:** A specific commit is checked out instead of a branch.
- **Solution:**
 1. Create a branch from the detached state: `git checkout -b <new-branch>`.
 2. Continue working or merge with another branch.

1.6 "fatal: remote origin already exists"

- **Cause:** Adding a remote repository that already exists.
- **Solution:**
 1. Update the existing remote: `git remote set-url origin <new-URL>`.

1.7 "Large file exceeds limit"

- **Cause:** Attempt to push a file exceeding the repository limit (e.g., GitHub's 100MB limit).
- **Solution:**
 1. Use Git Large File Storage (LFS): `git lfs track "<file-pattern>"`.

1.8 "Submodule update failed"

- **Cause:** Incorrect or inaccessible submodule configuration.
- **Solution:**
 1. Update submodules: `git submodule update --init --recursive`.

1.9 "fatal: cannot lock ref"

- **Cause:** A .lock file is preventing operations due to an interrupted process.
- **Solution:**
 1. Remove the .lock file: `rm -f .git/index.lock`.

1.10 "Untracked files prevent switching branches"

- **Cause:** Untracked changes conflict with the branch switch.
- **Solution:**
 1. Stash changes: `git stash`.

2. Switch branches: `git checkout <branch>`.

2. Jenkins Errors

2.1 "Jenkins service not starting"

- **Cause:** Corrupted configuration or missing Java installation.
- **Solution:**
 1. Check logs: `sudo journalctl -u jenkins`.
 2. Verify Java installation: `java -version`.

2.2 "Build stuck in the queue"

- **Cause:** No available executors or agents.
- **Solution:**
 1. Ensure the Jenkins agent is running and connected.
 2. Increase executor count under "Manage Jenkins."

2.3 "Plugins fail to load"

- **Cause:** Outdated Jenkins version or missing dependencies.
- **Solution:**
 1. Update Jenkins to the latest version.
 2. Reinstall failing plugins.

2.4 "Pipeline script syntax error"

- **Cause:** Incorrect Groovy syntax.
- **Solution:**
 1. Validate the script using the pipeline syntax generator.
 2. Debug syntax errors with Jenkins logs.

2.5 "Build fails due to missing environment variables"

- **Cause:** Environment variables not set in the job configuration.
- **Solution:**
 1. Define variables in "Build Environment" or as parameters.
 2. Use `env.VARIABLE_NAME` in the pipeline script.

2.6 "Unauthorized webhook trigger"

- **Cause:** Incorrect webhook configuration or missing credentials.
- **Solution:**
 1. Verify the webhook URL in the source control settings.
 2. Add proper credentials in Jenkins.

2.7 "Job not found after restart"

- **Cause:** Corrupted job configurations.
- **Solution:**
 1. Check for backups in `/var/lib/jenkins/jobs`.
 2. Recreate the job manually if backups are unavailable.

2.8 "Out of disk space"

- **Cause:** Jenkins workspace consumes too much space.
- **Solution:**
 1. Clean old builds: Manage Jenkins > Workspace Cleanup Plugin.
 2. Automate workspace cleanup post-build.

2.9 "Build artifacts not archiving"

- **Cause:** Incorrect artifact path configuration.
- **Solution:**
 1. Ensure correct file paths in "Archive the artifacts" step.

2.10 "Node disconnected unexpectedly"

- **Cause:** Network or agent-side issues.
- **Solution:**
 1. Verify agent logs and network connectivity.

3. Docker Errors

3.1 "Cannot connect to the Docker daemon"

- **Cause:** Docker service is not running or user lacks permissions.
- **Solution:**
 1. Start the Docker service: `sudo systemctl start docker`.
 2. Add the user to the Docker group: `sudo usermod -aG docker $USER`.

3.2 "Port is already in use"

- **Cause:** Another process is bound to the same port.
- **Solution:**
 1. Stop the conflicting container: `docker stop <container-id>`.
 2. Change the container's port mapping.

3.3 "No space left on device"

- **Cause:** Disk space is exhausted by Docker images and containers.
- **Solution:**
 1. Remove unused resources: `docker system prune -a`.

3.4 "ImagePullBackOff" in Kubernetes

- **Cause:** Invalid image tag or registry issues.
- **Solution:**
 1. Verify the image name and registry credentials.

3.5 "Permission denied on bind mount"

- **Cause:** Host directory permissions are restrictive.
- **Solution:**
 1. Update directory permissions: `chmod 777 <directory>`.

4. Kubernetes Errors

4.1 "ImagePullBackOff"

- **Cause:** The Kubernetes node cannot pull the specified container image due to an invalid image name, tag, or lack of access.
- **Solution:**
 1. Verify the image name and tag.
 2. If the image is private, check the ImagePullSecret or use `kubectrl create secret`.

4.2 "CrashLoopBackOff"

- **Cause:** The container crashes repeatedly, often due to application-level errors.
- **Solution:**
 1. Check container logs using `kubectrl logs <pod-name>`.
 2. Fix the underlying issue causing the crash.

4.3 "Node Not Ready"

- **Cause:** A Kubernetes node is unhealthy or cannot join the cluster.
- **Solution:**
 1. Use `kubectrl get nodes` to check node status.
 2. Restart the kubelet service and verify resource availability.

4.4 PersistentVolumeClaim Pending

- **Cause:** No PersistentVolume matches the claim's requirements.
- **Solution:**
 1. Check storage class and create a matching PersistentVolume.

4.5 "Pod is stuck in Pending state"

- **Cause:** Insufficient resources or unschedulable conditions.
- **Solution:**
 1. Check node capacity: `kubectrl describe pod <pod-name>`.

-
2. Scale up resources or adjust pod resource requests.

4.6 "RBAC: Access Denied"

- **Cause:** Role-based access control prevents the action.
- **Solution:**
 1. Grant proper permissions using Role/ClusterRole and RoleBinding.

4.7 Service Unreachable

- **Cause:** Service or ingress misconfiguration.
- **Solution:**
 1. Verify Service type, selectors, and target port.
 2. Check the ingress rules if applicable.

4.8 "Resource Quota Exceeded"

- **Cause:** The namespace has hit resource limits.
- **Solution:**
 1. Increase the resource quota or optimize usage.

4.9 "Evicted Pods"

- **Cause:** Nodes lack sufficient resources.
- **Solution:**
 1. Check events with `kubectl describe pod`.
 2. Add resources or reschedule workloads.

4.10 Deployment Rollout Fails

- **Cause:** Health checks fail for new pods.
- **Solution:**
 1. Check logs and events of failed pods.
 2. Roll back using `kubectl rollout undo`.

5. Ansible Errors

5.1 "Host unreachable"

- **Cause:** SSH connection fails.
- **Solution:**
 1. Verify SSH keys and access.
 2. Ensure the inventory file has the correct IP addresses.

5.2 "Syntax error in playbook"

- **Cause:** Incorrect YAML formatting.
- **Solution:**
 1. Validate YAML with a linter or `ansible-playbook --syntax-check`.

5.3 "Undefined variable"

- **Cause:** Variable is not defined in the playbook or inventory.
- **Solution:**
 1. Define the variable in `vars`, `group_vars`, or the inventory file.

5.4 "Command not found" on target node

- **Cause:** Required packages are missing.
- **Solution:**
 1. Install missing packages using a task in the playbook.

5.5 "Permission denied"

- **Cause:** User lacks required permissions on the target machine.
- **Solution:**
 1. Use `become: true` in the playbook to execute as a superuser.

5.6 Module not found

- **Cause:** The Ansible module is not installed.
- **Solution:**
 1. Ensure Ansible and its dependencies are up-to-date.

5.7 "Failed to find group_vars"

- **Cause:** Incorrect directory structure.
- **Solution:**
 1. Place group_vars and host_vars in the same directory as the inventory file.

5.8 Playbook runs indefinitely

- **Cause:** Task stuck in a loop or waiting for an unavailable service.
- **Solution:**
 1. Add timeout options and validate tasks.

5.9 "Handler not triggered"

- **Cause:** No task notifies the handler.
- **Solution:**
 1. Add notify: <handler-name> to the appropriate task.

5.10 Dynamic inventory script failure

- **Cause:** Errors in the inventory script.
- **Solution:**
 1. Debug the script manually or use --list to validate.

6. Terraform Errors

6.1 "Provider plugin not found"

- **Cause:** Missing or outdated provider plugin.
- **Solution:**
 1. Run terraform init to download required plugins.

6.2 "State file lock error"

- **Cause:** Concurrent Terraform runs.
- **Solution:**
 1. Unlock the state file: terraform force-unlock <lock-id>.

6.3 "Error acquiring the state lock"

- **Cause:** Network issues when using remote state.
- **Solution:**
 1. Retry after ensuring stable connectivity.

6.4 Resource already exists

- **Cause:** Attempting to create a resource that already exists.
- **Solution:**
 1. Use terraform import to manage existing resources.

6.5 "Plan does not match changes"

- **Cause:** Drift in the state file.
- **Solution:**
 1. Refresh state: terraform refresh.

6.6 "Invalid index" in Terraform output

- **Cause:** Incorrect usage of lists or maps.
- **Solution:**
 1. Validate variable types and indices.

6.7 "Module not found"

-
- **Cause:** Incorrect module path.
 - **Solution:**
 1. Ensure module paths are correct and run terraform get.

6.8 Authentication failure

- **Cause:** Missing or invalid credentials.
- **Solution:**
 1. Set proper environment variables or credentials files.

6.9 "Timeout waiting for resource"

- **Cause:** Resource creation takes too long.
- **Solution:**
 1. Increase timeout settings in the resource block.

6.10 "Remote backend configuration error"

- **Cause:** Incorrect backend configuration.
- **Solution:**
 1. Check and fix the backend block in the Terraform configuration.

7. Prometheus Errors

7.1 "No targets found"

- **Cause:** Prometheus is not configured to scrape any endpoints.
- **Solution:**
 1. Check prometheus.yml for scrape configurations.
 2. Ensure that target endpoints are reachable.

7.2 "Prometheus not scraping data"

- **Cause:** Target endpoints are misconfigured or not reachable.
- **Solution:**
 1. Verify the status of targets in the Prometheus UI under **Status > Targets**.
 2. Fix any endpoint issues or configurations.

7.3 "High cardinality in metrics"

- **Cause:** Too many unique label combinations in metrics.
- **Solution:**
 1. Optimize label usage in your metric definitions.
 2. Use relabeling rules to filter out unnecessary labels.

7.4 "Prometheus service not starting"

- **Cause:** Misconfiguration in prometheus.yml or insufficient resources.
- **Solution:**
 1. Validate the configuration file: `promtool check config prometheus.yml`.
 2. Check system resources and allocate more CPU/memory if needed.

7.5 "Query taking too long"

- **Cause:** Large datasets or inefficient queries.
- **Solution:**

1. Optimize your PromQL queries by limiting time ranges or labels.
2. Enable query caching for better performance.

7.6 "Prometheus out of storage space"

- **Cause:** Retention period or data volume exceeds available disk space.
- **Solution:**
 1. Reduce the data retention period: --
storage.tsdb.retention.time=<duration>.
 2. Add more storage or clean old data manually.

7.7 "Prometheus alert not firing"

- **Cause:** Misconfigured alert rules.
- **Solution:**
 1. Validate alert rules using promtool check rules.
 2. Ensure the alert expression is correct and matches expected data.

7.8 "Prometheus crash due to OOM (Out of Memory)"

- **Cause:** Too many metrics or insufficient memory allocation.
- **Solution:**
 1. Increase memory allocation to the Prometheus server.
 2. Reduce the number of metrics being scraped by filtering them.

7.9 "Scraped data is incomplete"

- **Cause:** Targets are partially down or network issues.
- **Solution:**
 1. Check the health of scrape targets.
 2. Monitor network connections between Prometheus and endpoints.

7.10 "Failed to reload configuration"

- **Cause:** Syntax errors in prometheus.yml.
- **Solution:**

-
1. Validate the configuration file with promtool.
 2. Fix any syntax issues before reloading.

8. ELK Stack Errors (Elasticsearch, Logstash, and Kibana)

8.1 Elasticsearch: "Cluster health is red"

- **Cause:** Some nodes or shards are unavailable.
- **Solution:**
 1. Check node status: `curl -X GET <ES_HOST>/_cat/nodes.`
 2. Reallocate shards: `POST _cluster/reroute.`

8.2 Elasticsearch: "Java heap space error"

- **Cause:** Insufficient heap memory for Elasticsearch.
- **Solution:**
 1. Increase heap size in `jvm.options`.
 2. Use `Xms` and `Xmx` values not exceeding 50% of available memory.

8.3 Logstash: "Pipeline aborted due to error"

- **Cause:** Syntax error or incorrect configuration in `logstash.conf`.
- **Solution:**
 1. Validate the configuration: `logstash -t -f logstash.conf.`
 2. Fix the error based on logs.

8.4 Logstash: "Connection refused to Elasticsearch"

- **Cause:** Incorrect Elasticsearch endpoint in Logstash configuration.
- **Solution:**
 1. Update output { `elasticsearch { hosts => ["<ES_HOST>"] }}`.

8.5 Kibana: "Kibana server is not ready yet"

- **Cause:** Elasticsearch cluster is not reachable.
- **Solution:**
 1. Verify Elasticsearch connectivity.
 2. Restart Kibana after Elasticsearch is ready.

8.6 "Index pattern not found in Kibana"

- **Cause:** No matching indices in Elasticsearch.
- **Solution:**
 1. Ensure data is being sent to Elasticsearch.
 2. Recreate the index pattern in Kibana.

8.7 "Logstash not processing logs"

- **Cause:** Input or filter misconfiguration.
- **Solution:**
 1. Check input sources and logs for issues.
 2. Validate filters with small datasets.

8.8 Elasticsearch: "Index not found"

- **Cause:** Querying an index that doesn't exist.
- **Solution:**
 1. Check available indices: `curl -X GET <ES_HOST>/_cat/indices`.

8.9 Kibana: "Dashboard is empty"

- **Cause:** Missing or misconfigured visualizations.
- **Solution:**
 1. Verify data sources for visualizations.
 2. Check time filters on the dashboard.

8.10 Logstash: "Filebeat logs not received"

- **Cause:** Incorrect Filebeat-to-Logstash configuration.
- **Solution:**
 1. Check Filebeat output.logstash configuration.
 2. Ensure the correct port and protocol.

9. AWS DevOps Tools Errors

9.1 EC2: "Instance not reachable"

- **Cause:** Incorrect security group or network ACL configuration.
- **Solution:**
 1. Verify inbound rules for SSH/HTTP access.
 2. Check VPC and subnet configuration.

9.2 S3: "Access Denied"

- **Cause:** Missing permissions for the S3 bucket.
- **Solution:**
 1. Update bucket policies to allow access.
 2. Attach proper IAM roles or policies.

9.3 CodePipeline: "Failed to deploy"

- **Cause:** Deployment stage error.
- **Solution:**
 1. Check deployment logs for details.
 2. Verify IAM permissions for CodePipeline and CodeDeploy.

9.4 RDS: "Cannot connect to the database"

- **Cause:** Firewall or networking issues.
- **Solution:**
 1. Add the correct inbound rules to the RDS security group.
 2. Check database credentials.

9.5 CloudFormation: "Stack creation failed"

- **Cause:** Misconfigured templates or resource dependencies.
- **Solution:**
 1. Review the stack events for specific errors.
 2. Validate the template: `aws cloudformation validate-template`.

9.6 IAM: "Policy not authorized"

- **Cause:** Insufficient permissions in the policy.
- **Solution:**
 1. Modify the policy to include the required actions.

9.7 ECS: "Task failed to start"

- **Cause:** Resource limitations or misconfigured task definitions.
- **Solution:**
 1. Check task logs in CloudWatch.
 2. Ensure sufficient memory and CPU allocation.

9.8 Lambda: "Execution failed"

- **Cause:** Errors in the Lambda function code.
- **Solution:**
 1. Review CloudWatch logs for detailed errors.
 2. Update the function code to fix issues.

9.9 Route 53: "DNS record not resolving"

- **Cause:** Incorrect record type or misconfigured TTL.
- **Solution:**
 1. Verify DNS records in the Route 53 console.
 2. Check domain delegation settings.

9.10 CloudWatch: "Metrics not visible"

- **Cause:** Missing or misconfigured monitoring agents.
- **Solution:**
 1. Ensure the CloudWatch agent is running.
 2. Verify configuration files.

10. Azure DevOps Tools Errors

10.1 Pipelines: "Build agent unavailable"

- **Cause:** The agent is offline or misconfigured.
- **Solution:**
 1. Restart the agent service.
 2. Verify agent registration in Azure DevOps.

10.2 "Resource group deployment failed"

- **Cause:** Incorrect ARM template.
- **Solution:**
 1. Validate the ARM template before deployment.

10.3 "Release failed in Azure DevOps"

- **Cause:** Deployment script errors.
- **Solution:**
 1. Check logs for failed tasks.
 2. Fix script errors and re-run.

10.4 Repos: "Merge conflict during pull request"

- **Cause:** Simultaneous changes to the same files.
- **Solution:**
 1. Resolve conflicts using the web editor or locally.

10.5 "Pipeline YAML syntax error"

- **Cause:** Incorrect YAML configuration.
- **Solution:**
 1. Use the Azure DevOps YAML validator.

10.6 "Artifacts not found"

- **Cause:** Missing build artifacts in the pipeline.
- **Solution:**

1. Ensure the correct artifact paths are defined in the build stage.

10.7 "Access denied to the Azure subscription"

- **Cause:** Missing role assignments.
- **Solution:**
 1. Assign proper roles to the service principal.

10.8 "Cannot connect to Azure Kubernetes Service (AKS)"

- **Cause:** Misconfigured kubeconfig or network rules.
- **Solution:**
 1. Regenerate kubeconfig using the Azure CLI: `az aks get-credentials`.

10.9 Boards: "Work item cannot be updated"

- **Cause:** Permission issues.
- **Solution:**
 1. Update user permissions in the project settings.

10.10 "Failed to publish test results"

- **Cause:** Incorrect test result path.
- **Solution:**
 1. Verify the test result files in the pipeline logs.

11. CI/CD Pipeline Errors

11.1 "Pipeline stuck in pending state"

- **Cause:** No available runners or agents.
- **Solution:**
 1. Ensure runners or agents are properly registered and running.
 2. Check tags and runner configurations to match the job requirements.

11.2 "Build fails due to dependency issues"

- **Cause:** Missing, outdated, or incorrect dependencies.
- **Solution:**
 1. Update dependency files like requirements.txt, package.json, or pom.xml.
 2. Cache dependencies to reduce build times and errors.

11.3 "Environment variable not found"

- **Cause:** Undefined or incorrectly set environment variables.
- **Solution:**
 1. Add the variable in the CI/CD tool's settings.
 2. Use .env files or secret managers for sensitive variables.

11.4 "Permission denied during deployment"

- **Cause:** User lacks required permissions on the target server.
- **Solution:**
 1. Ensure proper SSH keys or access tokens are configured.
 2. Use a service account with deployment privileges.

11.5 "Timeout in build or deployment stage"

- **Cause:** Long-running processes exceed the timeout threshold.
- **Solution:**
 1. Optimize pipeline stages for efficiency.

2. Increase timeout values in the pipeline configuration.

11.6 "Pipeline fails only on specific branches"

- **Cause:** Branch-specific settings or missing configurations.
- **Solution:**
 1. Verify branch-specific variables and scripts.
 2. Use conditional logic to apply configurations only for certain branches.

11.7 "Docker image not found in the pipeline"

- **Cause:** Incorrect image name, tag, or lack of authentication.
- **Solution:**
 1. Check and update the image name and tag.
 2. Authenticate with the Docker registry using proper credentials.

11.8 "Failed to upload artifacts"

- **Cause:** Incorrect artifact paths or permissions.
- **Solution:**
 1. Validate the artifact paths defined in the pipeline.
 2. Ensure proper permissions to upload files to storage services.

11.9 "Parallel jobs failing intermittently"

- **Cause:** Shared resources causing conflicts between parallel jobs.
- **Solution:**
 1. Use isolation mechanisms like resource locks.
 2. Clean up shared resources after each job.

11.10 "Webhook not triggering pipeline"

- **Cause:** Misconfigured webhook or restricted access.
- **Solution:**
 1. Verify the webhook URL and payload.
 2. Check firewall or network restrictions blocking the webhook.

12. Monitoring Tools Errors

12.1 "Metrics not visible in Grafana"

- **Cause:** Incorrect data source configuration or no metrics collected.
- **Solution:**
 1. Verify data source settings in Grafana.
 2. Check Prometheus targets or other metrics sources for proper data collection.

12.2 "Prometheus data missing from dashboards"

- **Cause:** Misconfigured PromQL queries or data source issues.
- **Solution:**
 1. Test PromQL queries in Prometheus before using them in Grafana.
 2. Check and fix Grafana queries for the correct metrics and labels.

12.3 "Logs not visible in ELK Stack"

- **Cause:** Filebeat, Logstash, or Elasticsearch misconfiguration.
- **Solution:**
 1. Check Filebeat logs for input/output errors.
 2. Validate Logstash pipeline and ensure Elasticsearch indices are configured correctly.

12.4 "High load on monitoring servers"

- **Cause:** Excessive data ingestion or high cardinality metrics.
- **Solution:**
 1. Optimize metrics collection by reducing labels and unnecessary data.
 2. Scale monitoring servers horizontally.

12.5 "Alerts not firing in Prometheus"

- **Cause:** Incorrect alert rules or expression issues.
- **Solution:**

1. Verify and test alert rules in Prometheus.
2. Use promtool to validate the alert configuration.

12.6 "Outdated Grafana dashboards"

- **Cause:** Data caching or lack of auto-refresh.
- **Solution:**
 1. Enable auto-refresh for dashboards.
 2. Clear cached data and reload the dashboard.

12.7 "Log ingestion delay in ELK Stack"

- **Cause:** Network bottlenecks or overloaded Logstash instances.
- **Solution:**
 1. Check network latency between Filebeat and Logstash.
 2. Scale Logstash horizontally or increase processing threads.

12.8 "Scraped data is incomplete"

- **Cause:** Partial scrape failures due to unreachable targets.
- **Solution:**
 1. Verify target status in Prometheus under **Status > Targets**.
 2. Fix any issues with endpoints or networking.

12.9 "Too many false-positive alerts"

- **Cause:** Overly sensitive alert thresholds.
- **Solution:**
 1. Adjust alert thresholds to reduce noise.
 2. Use deadband or hysteresis to filter transient anomalies.

12.10 "Dashboard panels showing no data"

- **Cause:** Incorrect queries or invalid time range.
- **Solution:**
 1. Verify the query syntax and metrics used in the panels.
 2. Ensure the selected time range matches the data availability.

Conclusion

The world of DevOps is inherently dynamic, with multiple tools and technologies working together to streamline software delivery and infrastructure management. Despite the power and flexibility of these tools, errors and issues are inevitable, especially in complex environments. This guide aims to provide a practical and comprehensive resource for DevOps practitioners to troubleshoot and resolve common errors across a wide array of tools, including **Git, Jenkins, Docker, Kubernetes, Ansible, Terraform, ELK Stack, AWS, Azure, and more.**

From configuration errors and syntax issues to network connectivity and resource limitations, the errors outlined in this guide represent challenges faced by DevOps teams daily. Each problem is paired with actionable solutions, empowering you to identify and fix issues efficiently, ensuring smoother operations and fewer disruptions.

The key to successful troubleshooting lies not only in resolving issues but in learning from them. By identifying patterns in failures, implementing proactive monitoring, and adhering to best practices, you can build more resilient and robust systems. DevOps is as much about fostering collaboration and automation as it is about continuous improvement and problem-solving.

As you navigate the evolving landscape of DevOps, remember that errors are opportunities to learn, optimize, and innovate. This guide is just a starting point, and as you gain experience, your understanding of these tools will deepen, enabling you to tackle even the most complex challenges with confidence.

Keep experimenting, keep learning, and most importantly, keep building. The road to DevOps mastery is paved with errors, but each error brings you one step closer to excellence.